

NyayaMitra AI

4-Member / 4-Day Dev Allocation Map & Project Plan

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1. Team Roles & Responsibilities

To complete the NyayaMitra AI prototype in 4 days, the work is divided among 4 specialized roles. This ensures parallel development of the UI, backend logic, RAG engine, and presentation assets.

Member 1 *Lead Developer (Backend & RAG Core)*
Role Focus: FastAPI setup, local laws vector search database creation, chat endpoint logic.

Member 2 *Frontend Specialist (UI/UX Design)*
Role Focus: Vite-React dashboard shell, document highlights overlay, interactive timelines, checklist layouts.

Member 3 *AI & Integration Developer*
Role Focus: Gemini OCR script integration, Structured JSON prompt parsing, Dynamic Telugu translations.

Member 4 *QA, Accessibility & Presentation Lead*
Role Focus: Browser Text-to-Speech widgets, calendar event downloads, mock SMS/WhatsApp structures, slide deck compilation.

2. Day-by-Day Task Allocation Map

Below is the day-by-day task checklist for each of the 4 team members to maintain continuous deployment and validation.

DAY 1: Project Setup & Baseline Pipelines	
Team Member	Tasks & Deliverables
Member 1 (Backend)	- Initialize git repository and project directory structure. - Set up the FastAPI backend environment (requirements.txt, FastAPI init). - Write standard API response envelopes, CORS headers, and basic health checks.
Member 2 (Frontend)	- Create React-Vite project with TypeScript framework. - Setup project colors, layout grid, global styles (Deep Navy / Warm Gold accent theme). - Code the basic dashboard UI shell and folder routing.
Member 3 (AI/OCR)	- Implement the backend Gemini API integration wrapper. - Build the multi-format file uploader endpoint and connect it to Gemini for multimodal text extraction. - Write tests with sample PNG/PDF notices to verify raw OCR text extraction.
Member 4 (QA/Docs)	- Gather standard legal documents (eviction notices, lease agreements, consumer default orders). - Draft a detailed manual test sheet mapping expected outputs. - Create the IEEE hackathon presentation slides outline.

DAY 2: Knowledge Grounding & Classifier Pipeline	
Team Member	Tasks & Deliverables
Member 1 (Backend)	- Build 'laws_kb.json' containing structured Indian Civil, Tenant, and Consumer Protection laws. - Write the pure-Python cosine-similarity database index based on TF-IDF or text vectors. - Setup the vector query API endpoint (`/api/search-laws`).
Member 2 (Frontend)	- Build a beautiful file upload zone supporting drag-and-drop with clean progress states. - Design the dual-pane workspace layout (viewer on left, cards tabs on right). - Design placeholders for timelines and next steps cards.
Member 3 (AI/OCR)	- Define the structured JSON output validation schemas (using Pydantic models). - Construct prompts forcing Gemini to classify document types and retrieve legal reference IDs. - Ground the LLM output with retrieved RAG laws to compile plain summary translations.
Member 4 (QA/Docs)	- Configure automated Python test script suite using `pytest` for backend API integration. - Test backend OCR + retrieval pipelines using raw PDF notices and report results. - Prepare diagrams describing user interactions and database design for slides.

DAY 3: Workspace Views & Contextual Chat Assistant

Team Member	Tasks & Deliverables
Member 1 (Backend)	- Code backend endpoint `/api/chat` with session memory for follow-up document queries. - Implement backend formatting template to output a pre-drafted legal response draft (.txt format). - Write unit tests testing chatbot session states.
Member 2 (Frontend)	- Integrate notice timelines, milestone countdown blocks, and actionable checkboxes in the UI. - Build the tabbed panels: Summary, Legal References, and 'Take Action' layout views. - Create the chatbot interface layout overlay in the right workspace pane.
Member 3 (AI/OCR)	- Connect chatbot API query to RAG database context and upload document history. - Create the translation endpoint `/api/translate` mapping plain summary and actions to Telugu. - Verify translations maintain semantic legal meanings.
Member 4 (QA/Docs)	- Execute continuous verification loops checking interface data connections. - Design detailed visual slide aids showing offline support and WhatsApp channel mockups. - Write user manual guide files explaining how to navigate the dashboard.

DAY 4: Accessibility Integration, Polish & Presentation

Team Member	Tasks & Deliverables
Member 1 (Backend)	- Build the calendar schedule helper endpoint generating custom `.ics` file downloads. - Optimize response latency, cache repeated LLM outputs, and complete code commenting.
Member 2 (Frontend)	- Integrate browser Web Speech Synthesis controls (play, pause, rate control, language voice selection). - Add beautiful CSS hover micro-animations, glassmorphic UI cards, and connect translation switches. - Test layout compatibility across multiple screen sizes.
Member 3 (AI/OCR)	- Verify Gemini prompting accuracy and fine-tune TTS readouts to avoid robotic pacing. - Run load stress testing on the backend server. - Assist backend in deploying system locally.
Member 4 (QA/Docs)	- Verify multi-browser Speech synthesis playback controls and Telugu readout flows. - Complete README installation docs and project video demo recording as backup. - Rehearse the live pitch deck delivery and coordinate the slide timeline.

3. Core Milestones & Success Criteria

- **Day 1 Milestone:** Clean extracted legal notices and metadata returned within 4 seconds.
- **Day 2 Milestone:** A stable RAG grounding returning 3+ accurate legal sections matched to the notice.
- **Day 3 Milestone:** A fully connected dual-pane React UI displaying working timeline countdowns.
- **Day 4 Milestone:** Perfect Telugu audio playback and 1-click Google Calendar syncing verified.